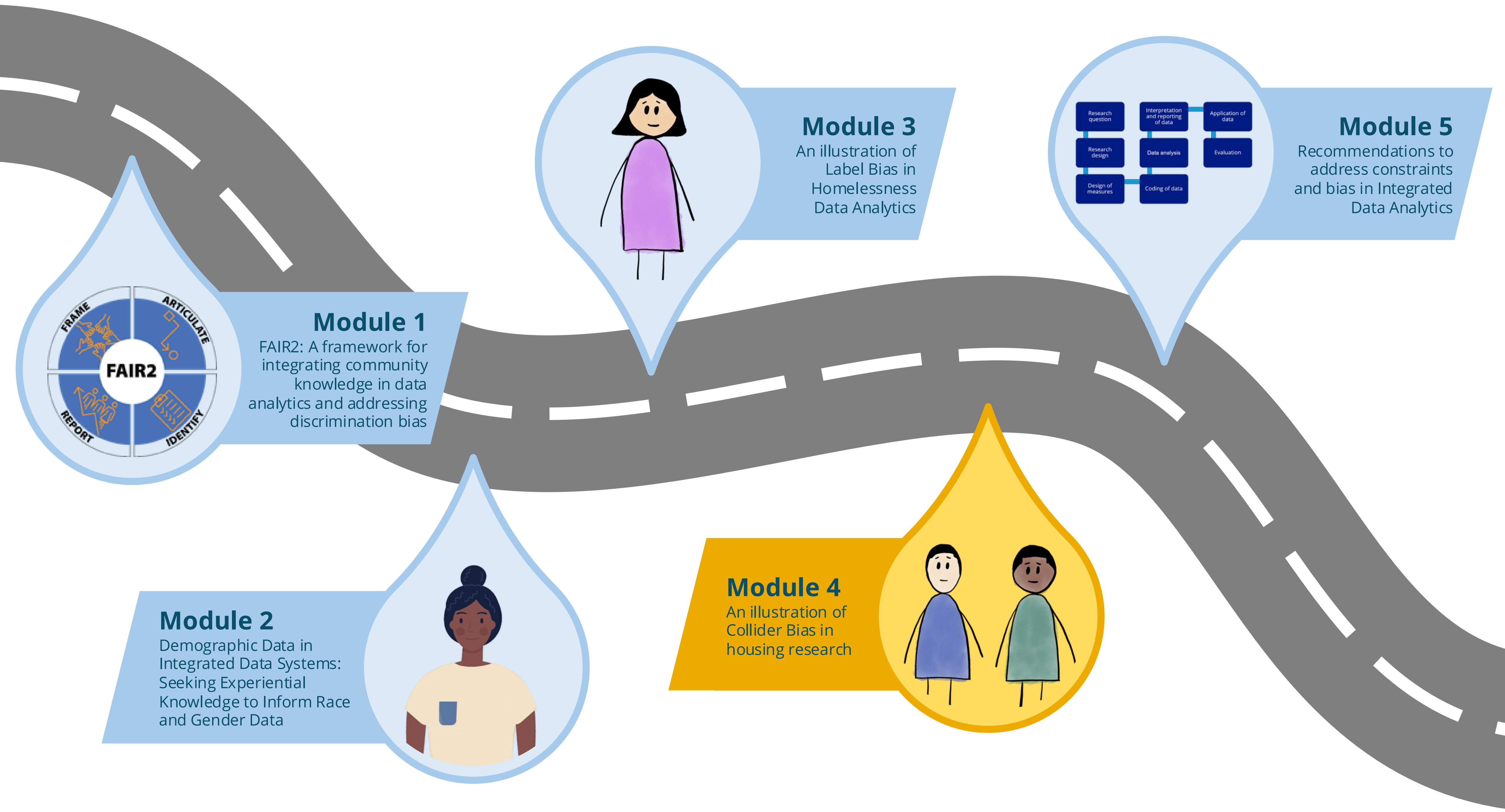
An illustration of Collider Bias in housing research

Centering the Public Interest in Integrated Administrative Data Analytics

Module 4 of 5

Public Interest Technology University Network Year 6 Challenge
January 2026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Amari and Liam have grown up together. Over the years, they've seen a lot of families move to more affluent areas. For Amari, his mom has had to work very hard for all they have and doesn't have extra cash. For Liam, his parents have had to spend a lot of money on healthcare bills for his dad's chronic health conditions. Neither Amari's nor Liam's families can afford to move.

As many of the White families have moved out, the neighborhood is now primarily Black except for families like Liam's. Amari's mom has given him many talks about growing up as a Black boy, and she's always careful about where she allows Amari to go.

This type of awareness is normal for Amari. In fact, many of his friends at school have also been given talks about growing up Black in this world. Liam has struggled with his own family problems.

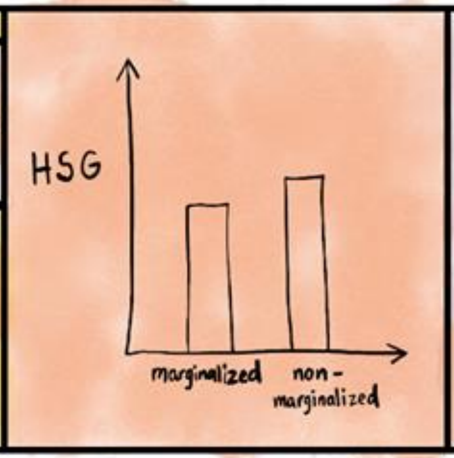
Although for different reasons, Amari and Liam have had bad experiences in school. Resources are scarce, teachers are overworked, and students are now more aware of how they are misperceived by society.

Amari and Liam are struggling with their grades. Although they are friends, neither feels that the other will understand the weight they carry inside.

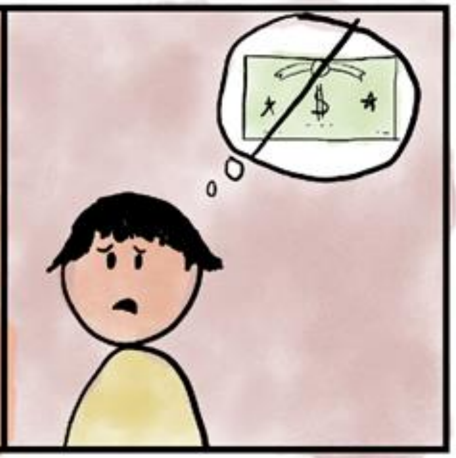
Harper is a community practice social worker who works with families in Amari and Liam's neighborhood. She's heard about the experiences of discrimination these families encounter and wonders how it affects academic performance for the kids.

Harper decides to use marginalization based on race as an indicator of discrimination. She pulls high school graduation rates from school data and race from family data from the city's income assistance program.

She's hoping to see how discrimination impacts high school graduation rates so she can apply for funding for a peer mentor program. This program matches students in the neighborhood with students in college who have shared identities.



However, when Harper looks at the data, she sees that among kids in Amari and Liam's neighborhood, marginalization based on race *barely* impacts graduation rates and is not statistically significant!

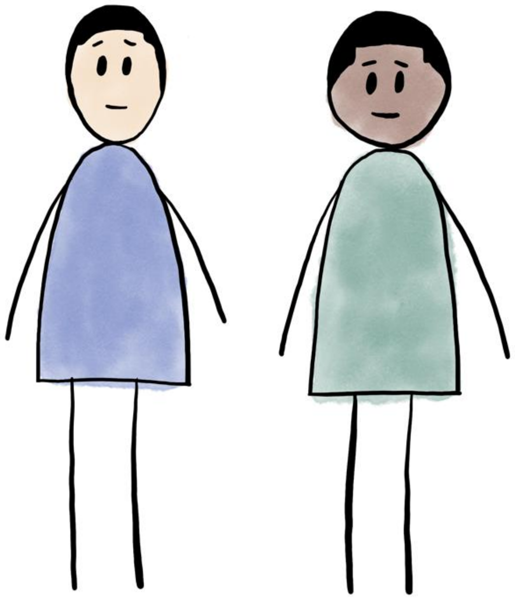


Harper concluded, "Well, this data won't be helpful in applying for that grant…"

Amari and Liam's story shows us that there are multiple reasons why people may end up in a particular population, like a neighborhood. Structural discrimination has limited opportunities for families like Amari's across generations. Liam's family experienced chronic health issues that took time, money, and resources. As such, neither Amari's nor Liam's families can afford to move to a neighborhood with more opportunities.

When looking ***within a particular population****, it's important to understand what factors contribute to people being in that population.*

Remember what Harper saw when looking <u>only at Amari and Liam's neighborhood</u>: No effect of discrimination on the likelihood of high school graduation.



What factors may contribute to Harper seeing little difference in high school graduation rates by race (-based marginalization) in Amari and Liam's neighborhood?

Introduction

You're a social worker in community practice, who primarily works with families in low-income neighborhoods. You notice that marginalization is one of the challenges these families and their neighborhoods face. You wonder how marginalization affects children as they are growing up and preparing to go out on their own. You ask yourself:

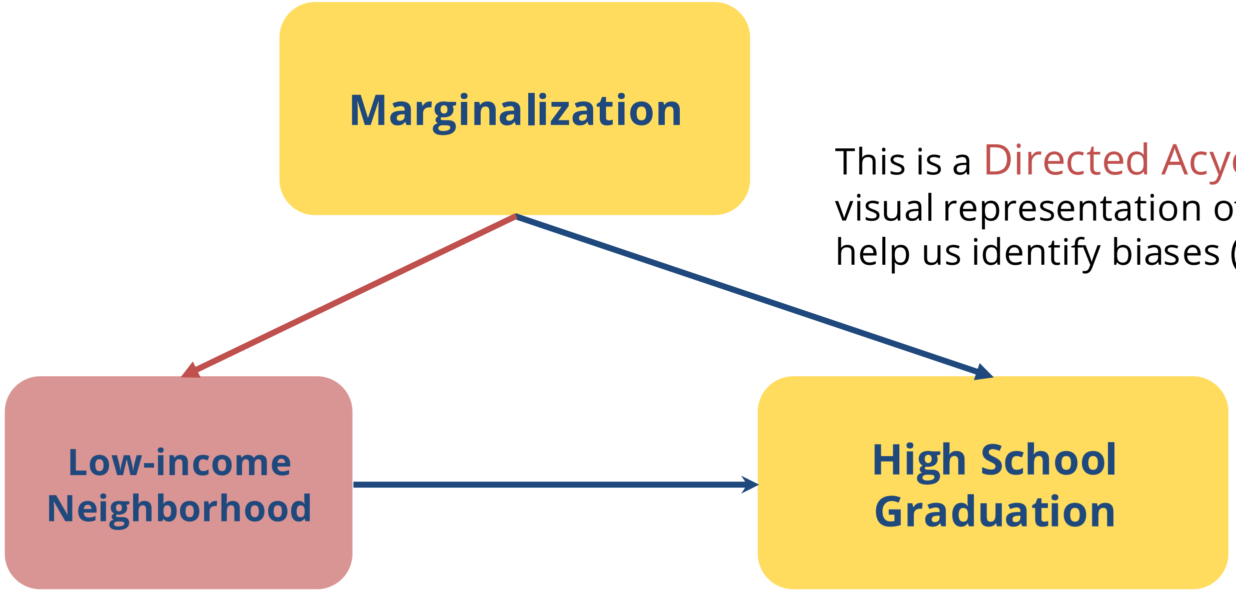
What are the effects of marginalization on the likelihood of high school graduation?

You received school data for these low-income neighborhoods including whether students graduate high school.

Let's think about how you're going to use that data to answer your questions…

Collider Bias in Research

- What are the effects of **marginalization on the likelihood of high school graduation**?
- You only have data on the individuals in the neighborhoods where you work
- Given the history of discrimination in housing policy, you know that marginalization also influences **neighborhood access**
- Let's start by building a very simple graphical model:

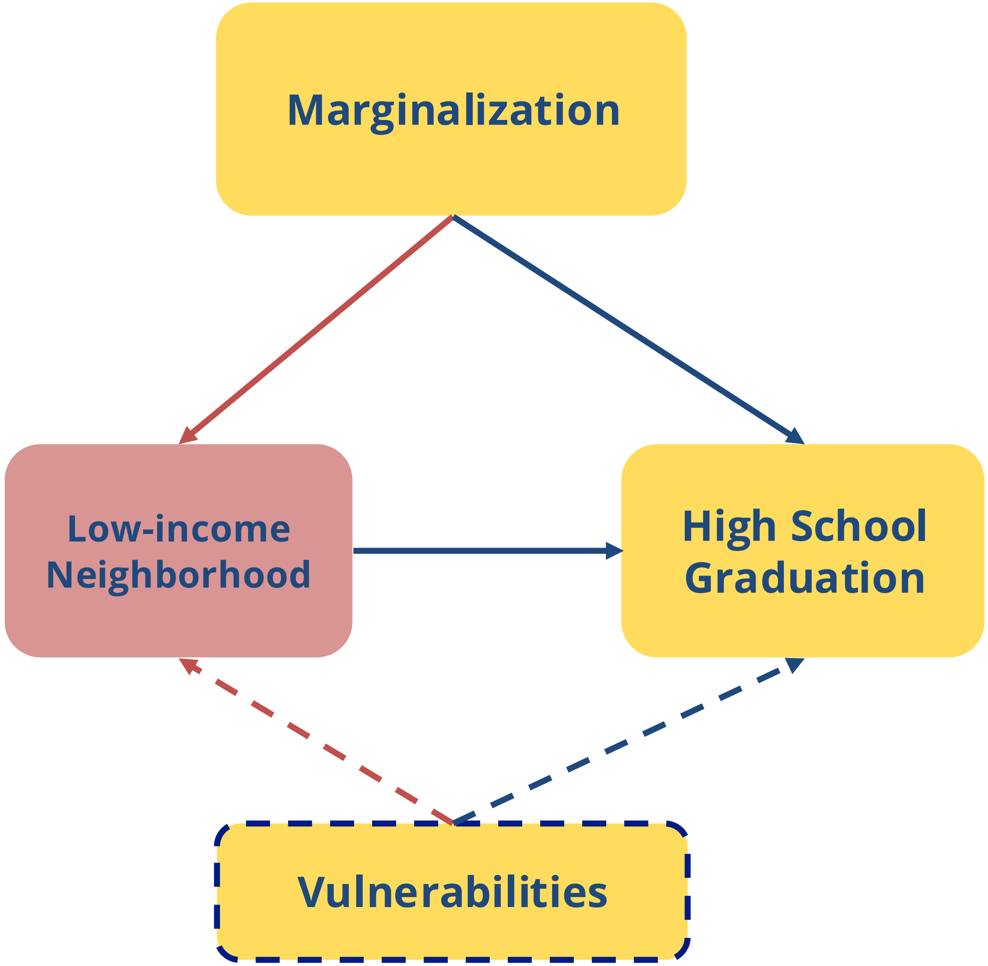


This is a Directed Acyclic Graph or **DAG**, a powerful visual representation of model assumptions that can help us identify biases (Pearl & Mackenzie, 2018).

Collider Bias in Research

- A more realistic graph recognizes that factors other than discrimination may lead to living in **low-income and low-resourced neighborhoods**
- We'll refer to these factors as vulnerabilities
- Vulnerabilities *may include health issues not recorded in the data* but still impacting the outcomes of interest

- **Consider Amari and Liam's story. What are some possible examples of vulnerabilities?**



Collider Bias in Research

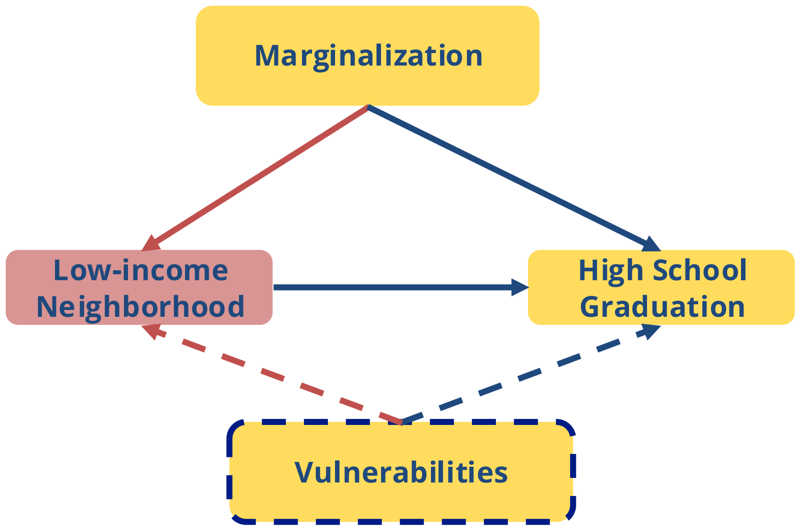
- Collider bias occurs when selecting on a **collider variable** (Elwert & Winship, 2014)
- See how two causal arrows point to low-income neighborhood?
  - A **collider variable** is a variable that is caused by two other correlated variables (i.e., marginalization and vulnerabilities)
  - If we only have data for schools in low-income neighborhoods, we are selecting on a collider variable:

Marginalization → Low-income Neighborhood ← Vulnerabilities

- This opens a non-causal pathway from marginalization to high school graduation…

Marginalization → Low-income Neighborhood ← Vulnerabilities → High School Graduation

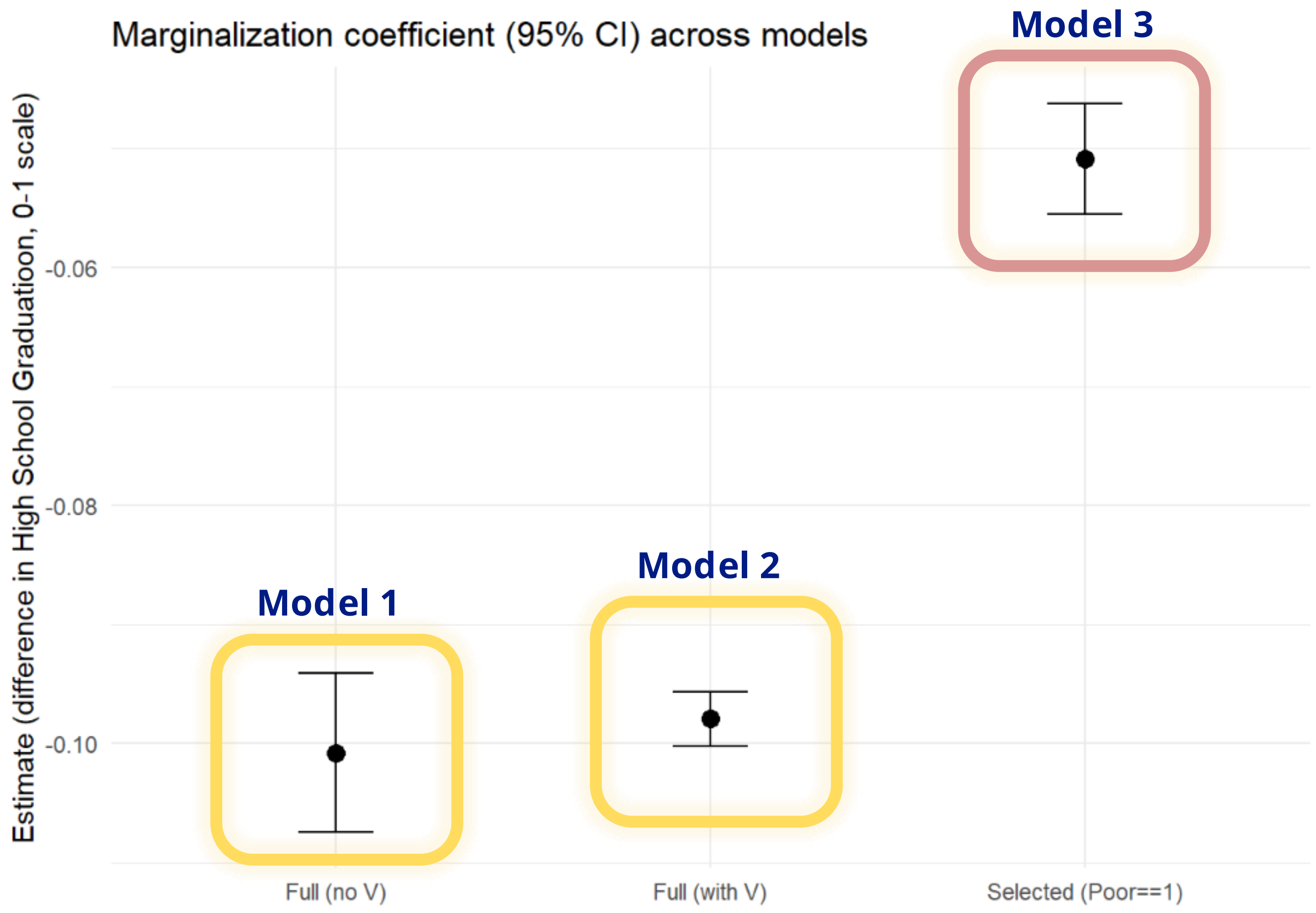
- ….and biases our causal estimate of marginalization on likelihood of high school graduation



Check out the R module for a simulated data example of how collider bias impacts our estimates of marginalization on high school graduation!

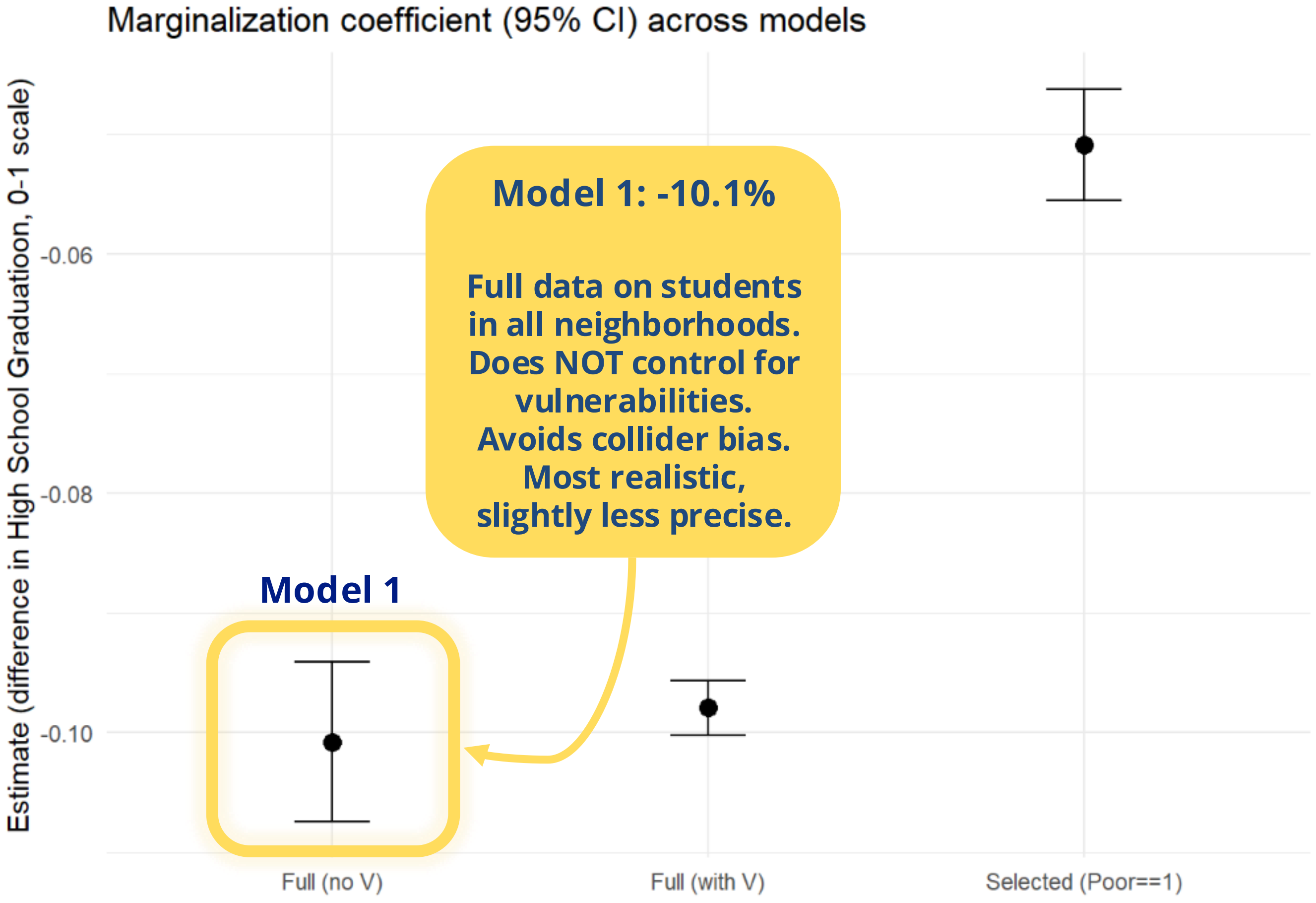
Simulated Data Example

- To explore the effect of marginalization on likelihood of graduation, we use four variables:
  - Marginalization
  - Vulnerabilities
  - Low-income Neighborhood
  - High school graduation
- Data is simulated in **three models**
- All y-axis numbers are negative, reflecting a negative impact in likelihood of graduation



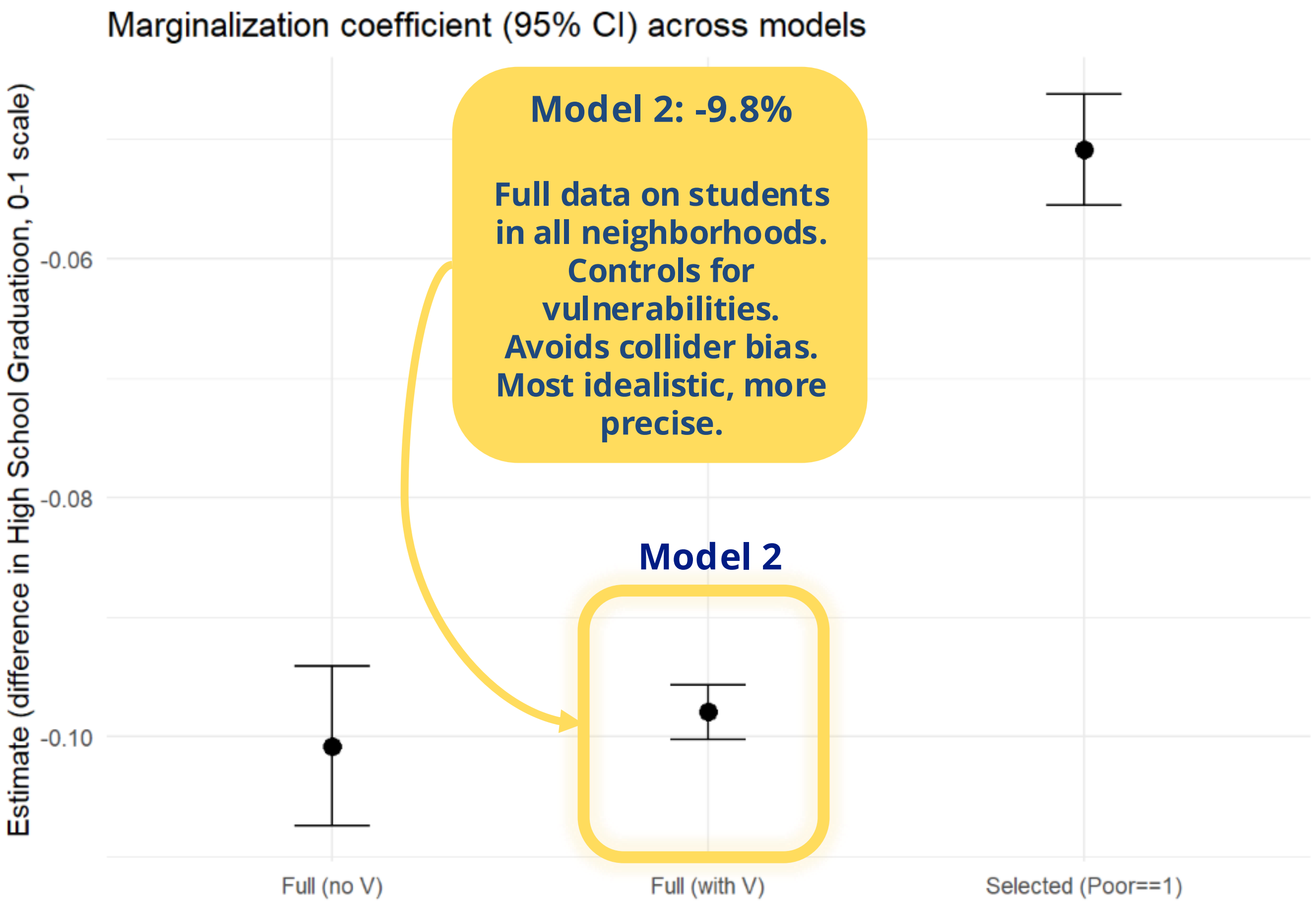
Simulated Data Example

- **Model 1**: Full data (no vulnerabilities)
  - Estimated impact of marginalization on likelihood of high school graduation
  - Uses full data (data on students in all neighborhoods)
  - Does NOT control for other variables (vulnerabilities)
    - Vulnerabilities not often captured in data
      - Most realistic, slightly less precise
    - Avoids collider bias
  - Effect on high school graduation: -10.1%
    - Conclusion: **Marginalization decreases likelihood of high school graduation by 10.1%**



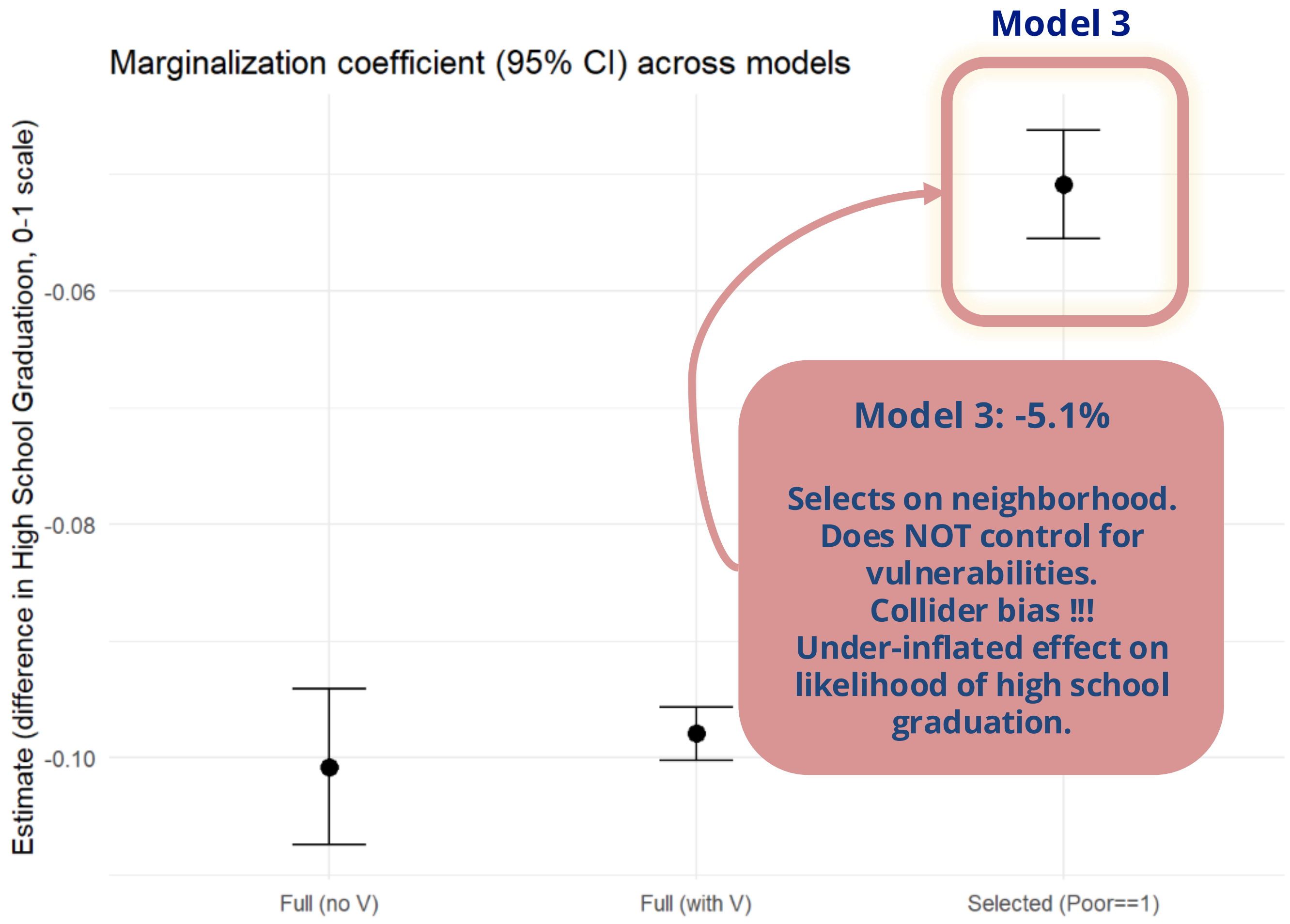
Simulated Data Example

- **Model 2**: Full data
  - Estimated impact of marginalization on likelihood of high school graduation
  - Uses full data (data on students in all neighborhoods)
  - Controls for other variables (vulnerabilities)
    - Increased precision but less realistic
    - Avoids collider bias
  - Effect on high school graduation: -9.8%
    - Conclusion: **Marginalization decreases likelihood of high school graduation by 9.8%**

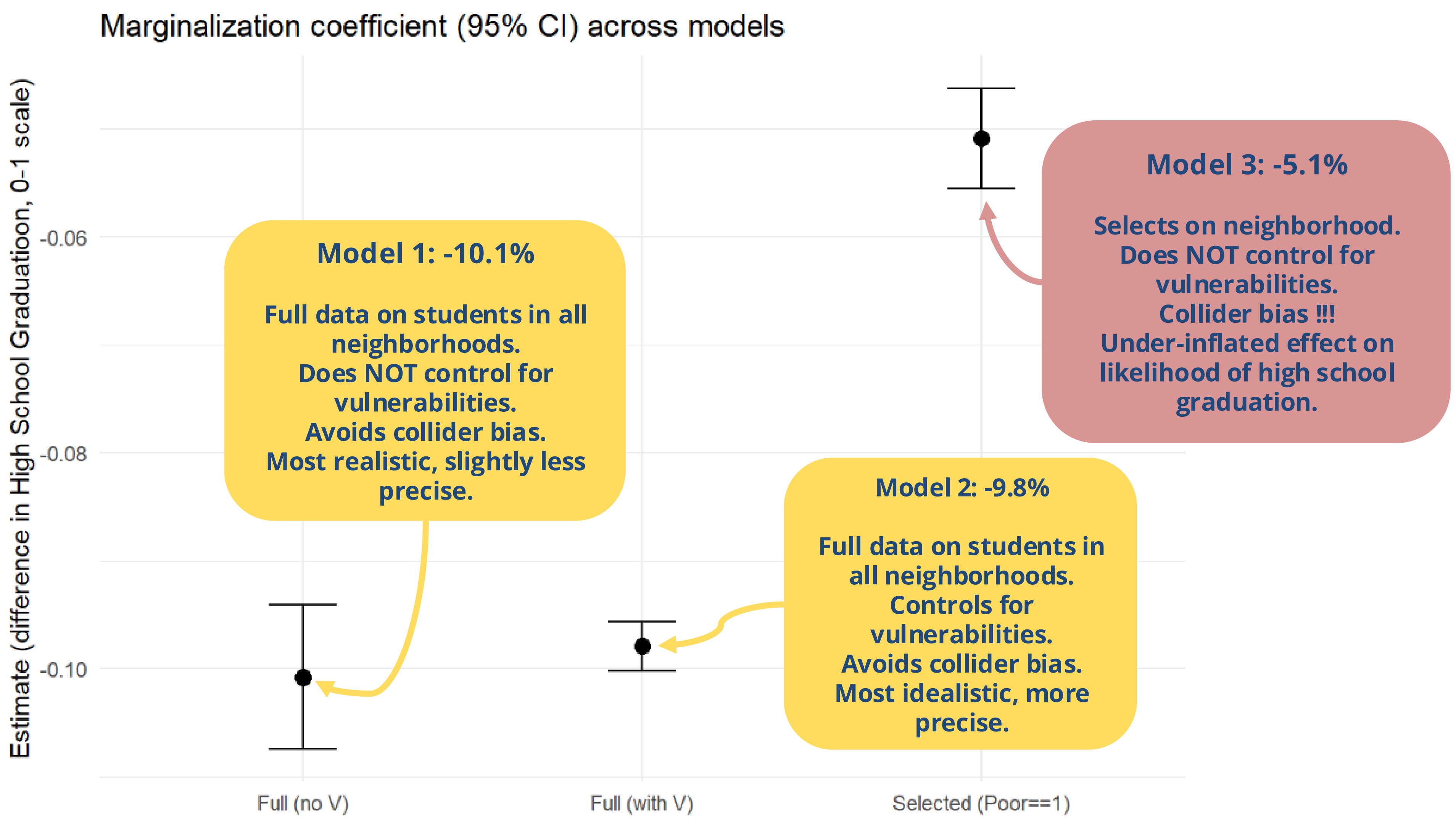


Simulated Data Example

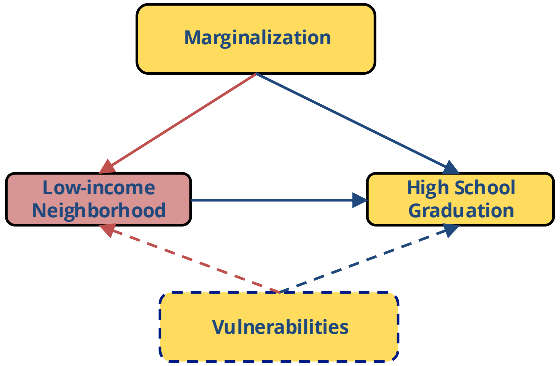
- **Model 3**: Harper's data
  - Estimated impact of marginalization on likelihood of high school graduation
  - Selects on neighborhood (only uses data for students in low-income neighborhoods)
    - Collider bias!
  - Does NOT control for other variables (vulnerabilities)
  - Effect on high school graduation: -5.1%
    - Conclusion: **Marginalization decreases likelihood of high school graduation by 5.1%**
    - Biased impact of marginalization on high school graduation, wrongly suggesting a small impact



Estimates of the impact of marginalization on the chance of graduating high school (0-1 scale) for models with and without collider bias



Consequences of Ignoring Collider Bias



The example highlights three main points:

1. Marginalization due to race and vulnerabilities (e.g. chronic health conditions in the family) are unrelated overall. Note: our DAG has no arrows *directly* connecting marginalization and vulnerabilities

- Marginalized people tend to live in poor neighborhoods because they're marginalized
- Non-marginalized people tend to live in low-income neighborhoods because of their vulnerabilities (e.g., chronic health conditions)
- In other words, marginalized & non-marginalized students tend to live in low-income neighborhoods for different, unrelated reasons

2. If we want to estimate the effects of marginalization on educational outcomes, but we only have information about students living in/attending school in low-income neighborhoods, in effect we're **comparing marginalized students to a subset of non-marginalized students with above-average levels of vulnerability**

3. This will **underestimate the negative effect of marginalization** on likelihood of high school graduation

Consequences of Ignoring Collider Bias

- We reach inaccurate conclusions that tend to **underestimate the impact of structural problems** in society
- This can negatively impact decision-making in social investments, funding allocation, policy-making, etc.
- It can help perpetuate **social biases**
  - A study using administrative police records found **no racial differences in police use of force** during encounters with civilians (Bui & Cox, 2016)
  - Another study later **proved this conclusion wrong**, showing how collider bias was responsible for these biased results (Knox et al., 2020)

What to do about it?

- **Diagnose with a directed acyclic graph (DAG)**: before engaging in data analytics, draw your assumptions about what factors influence your variables of interest and target population
    - Systemic factors (i.e. racism)
    - Individual-level factors (i.e. (dis)ability status)

- Note that using administrative data require us to think hard about whether collider bias might be lurking
- Administrative data may be population-level data, such as all students in a school district or all people seeking homeless services. However, it is NOT a random sample of the population at large
- Back to our story, Harper, the social worker, only had school data for Amari and Liam's neighborhood. This data **does not include other suburbs more accessible to non-marginalized students**, where disparities emerge
    - Could Harper access other school district data?

What to do about it?

Be transparent about the limitations of research.

Harper's Realization:

- **I can't conclude that discrimination doesn't matter**.
- I can only conclude that *for this specific neighborhood*, high poverty affects everyone's educational opportunities.
- This likely makes the effect of discrimination look smaller than it really is.
- I need county-wide school data to see the real pattern.

References

Elwert, F., & Winship, C. (2014). Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable. *Annual review of sociology*, 40, 31–53. https://doi.org/10.1146/annurev-soc-071913-043455

Hernández-Díaz, S., Schisterman, E. F., & Hernán, M. A. (2006). The birth weight "paradox" uncovered?. *American journal of epidemiology*, 164(11), 1115–1120. https://doi.org/10.1093/aje/kwj275

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Bui, Q., & Cox, A. (2016, July 11). *Surprising new evidence shows bias in police use of force but not in shootings.* The New York Times. https://www.nytimes.com/2016/07/12/upshot/surprising-new-evidence-shows-bias-in-police-use-of-force-but-not-in-shootings.html

Appendix: More Examples of Collider Bias

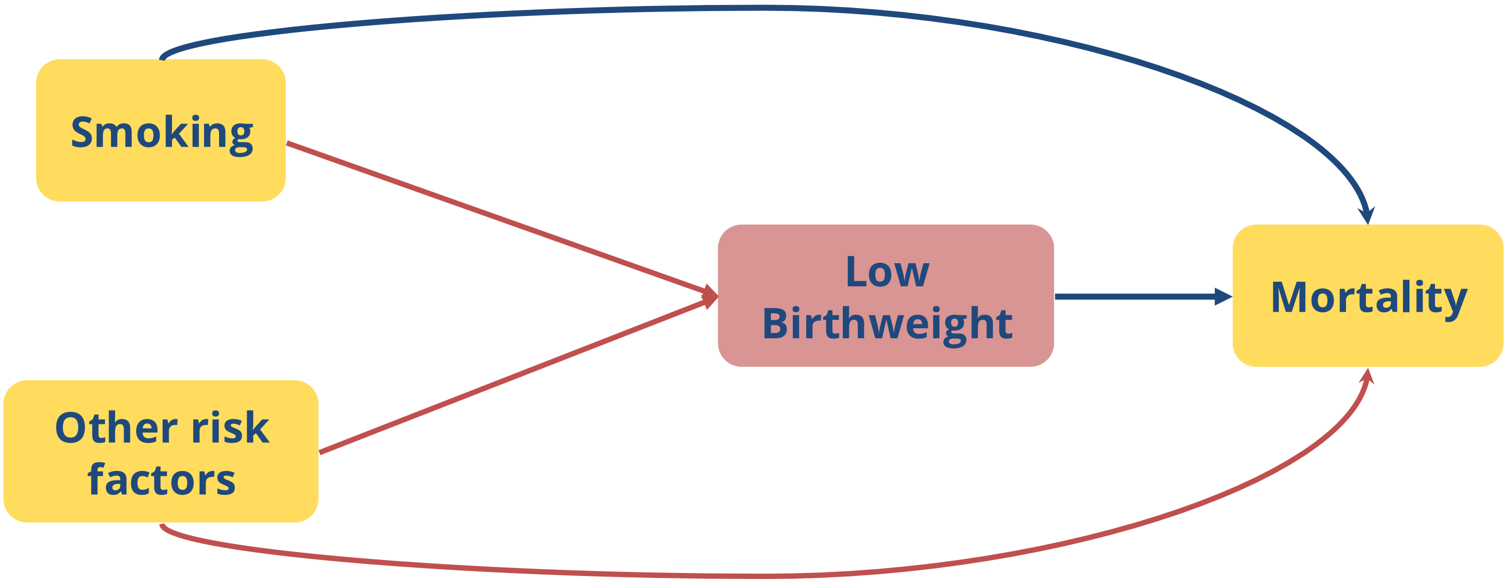
Low Birthweight Paradox

- Low birthweight is a risk factor for infant mortality (Low Birthweight → Mortality)
- Smoking is a risk factor for low birthweight (Smoking → Low Birthweight → Mortality)

- However: **among infants of low birthweight**, infant mortality is lower for infants born to parents who smoke (!)
- **What!? Why?**
  - One theory in the 1960s was that **parental smoking is beneficial to infants who happen to be born with low birthweight…** (Pearl & Mackenzie, 2018)
  - This paradox was not reliably explained until 2006 (Hernandez-Dıaz et al., 2006)

Collider Bias

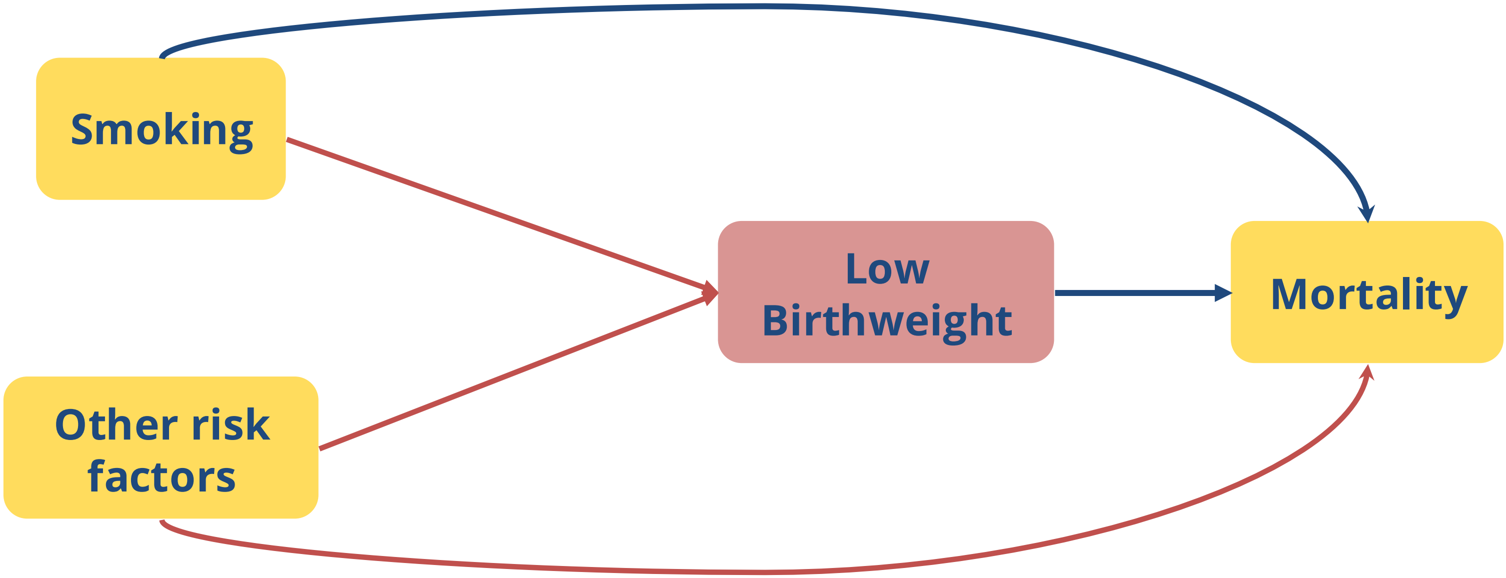
- The Birthweight Paradox appears because of **selection on a collider**:
- **Among infants of low birthweight (collider)**, infant mortality is **lower** for infants born to parents who smoke relative to other infants …



- What may cause infants to have low birthweight?
- What else contributes to low birthweight?

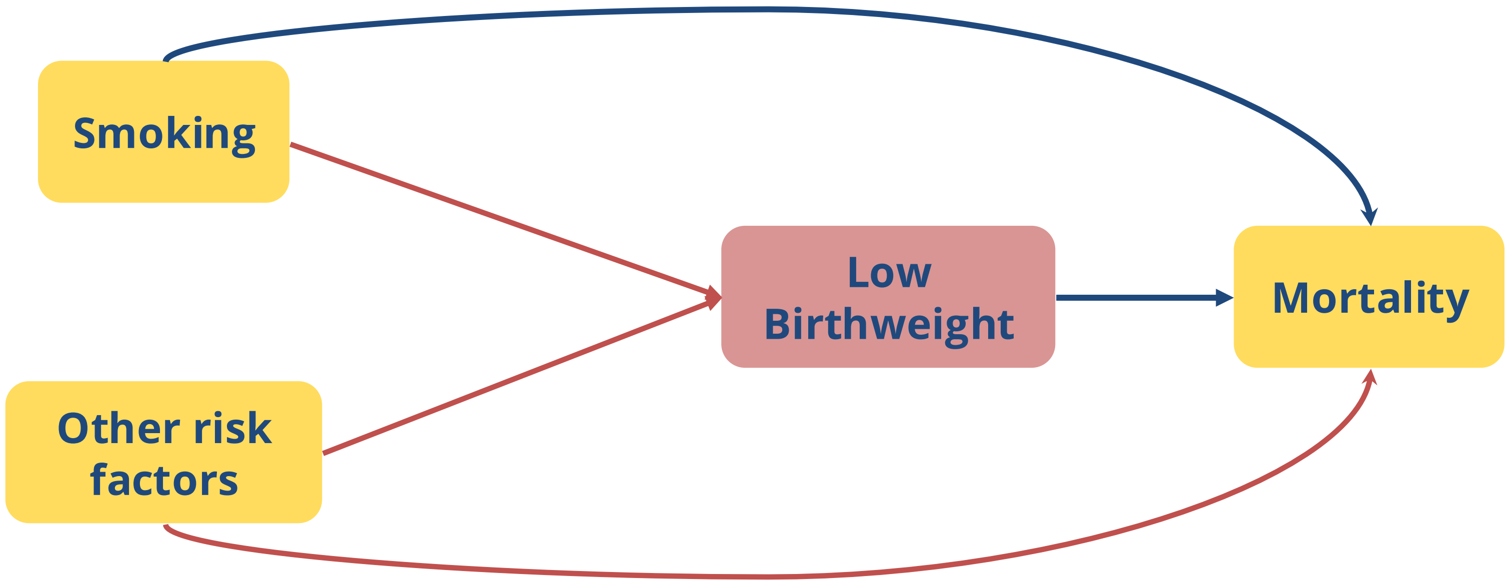
Understanding the Low Birthweight Paradox

- **A non-causal pathway is opened** when selecting on **low birthweight (collider variable),** biasing estimates of the causal effects of smoking on mortality
  - **Smoking → Low Birthweight ← Other risk factors → Mortality**
- **Other risk factors** (i.e., congenital diagnoses) also lead to low birthweight and are **high risk factors for infant mortality**!
- Smoking and other risk factors of low birthweight are uncorrelated in the general population but appear negatively correlated in the low birthweight sample
- Our low birthweight population is **NOT representative of all births**, so it is not suitable to study the impact of smoking on infant mortality



Low Birthweight Paradox

- Without realizing that low birthweight is a **collider variable**, we may **inaccurately** assume that **parental smoking is protective** for infants of low birthweight
- When we see low birthweight as a collider variable, we **more accurately understand who comprises our low birthweight population**

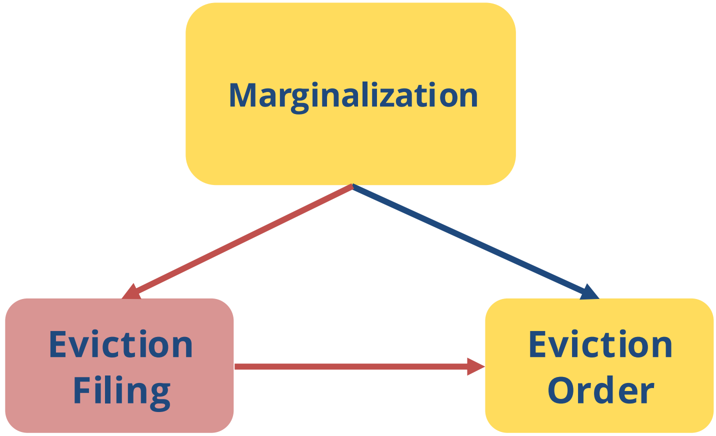


Collider Bias in Eviction Research

- In a study in Cleveland, OH, by Richter et al. (2021), a **positive association** was found between **White identity and receiving an eviction order** **among those who received an eviction filling**
  - Are White individuals more likely to receive an eviction order than individuals of a marginalized race?

Variables	Coefficent	SE
Previous case in past year	0.326***	(0.0423)
Male	0.161***	(0.0464)
Race: Other, unknown	− 0.0721	(0.0981)
Race: White	0.143***	(0.0447)
Age at filing: 25–39	− 0.0173	(0.0536)
Age at filing: 40–59	− 0.0719	(0.0602)
Age at filing: 60+	0.0913	(0.103)
Has children	0.0855**	(0.0409)
ADI above 80th percentile	0.0127	(0.0493)
Landlord: Nonprofit, govt.	0.259***	(0.0951)
Landlord: Corporate	0.710***	(0.0519)
Landlord: First–Last name	0.734***	(0.0493)
Year = 2015	− 0.0258	(0.0415)
Year = 2016	0.0471	(0.0435)
Constant	− 1.028***	(0.0799)
Number of observations	13,751	

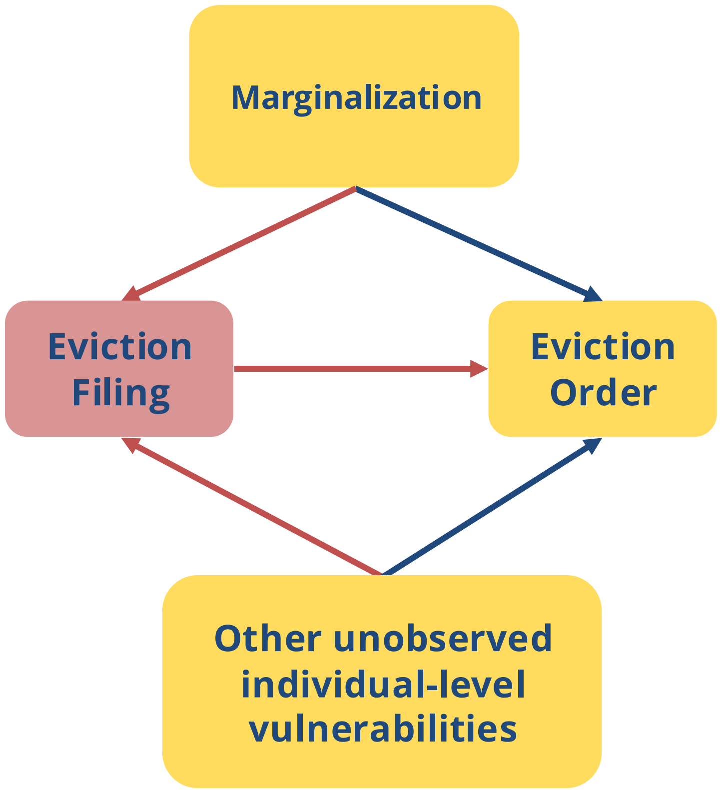
Note. Standard errors are given in parentheses.
**p* < .1. ***p* < .05. ****p* < .01.



Logistic regression model estimation of the likelihood of receiving an eviction order. The data represent *eviction case filings in the City of Cleveland, Ohio, 2013–2016, linked to public assistance records*. Variables correspond to the head of household associated with the eviction filing.

Collider Bias in Eviction Research

- Who makes up the **population of people who receive an eviction filing (collider variable)**?
- Marginalization plays a role in housing instability, but other vulnerabilities unrelated to marginalization, such as health issues may also contribute
- Selecting on eviction fillings (using this data in the analysis) cannot help us understand the impact of marginalization on receiving an eviction order
- We find that **among those who receive an eviction filing**, those identified as White are more likely to receive an eviction order, but …

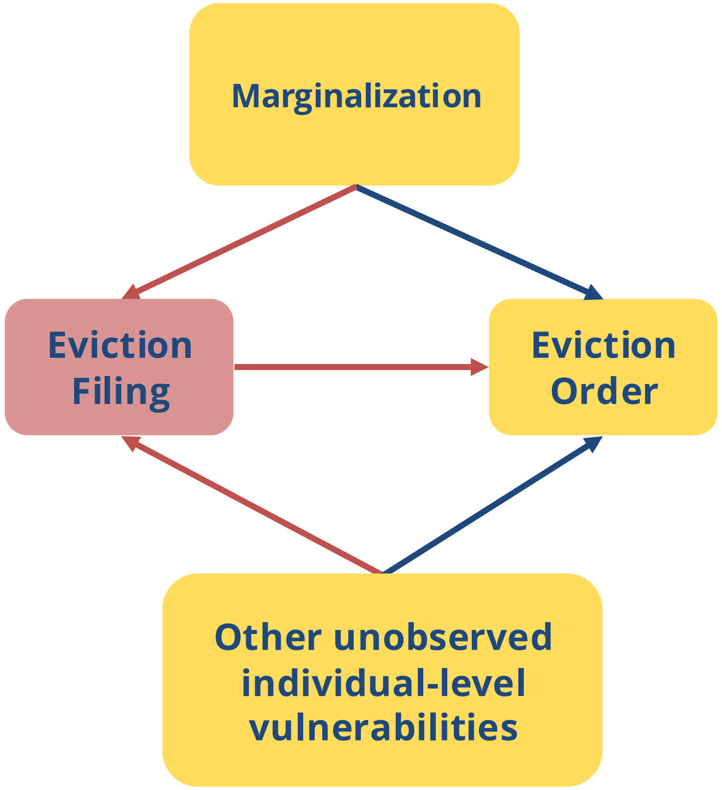


Collider Bias in Eviction Research

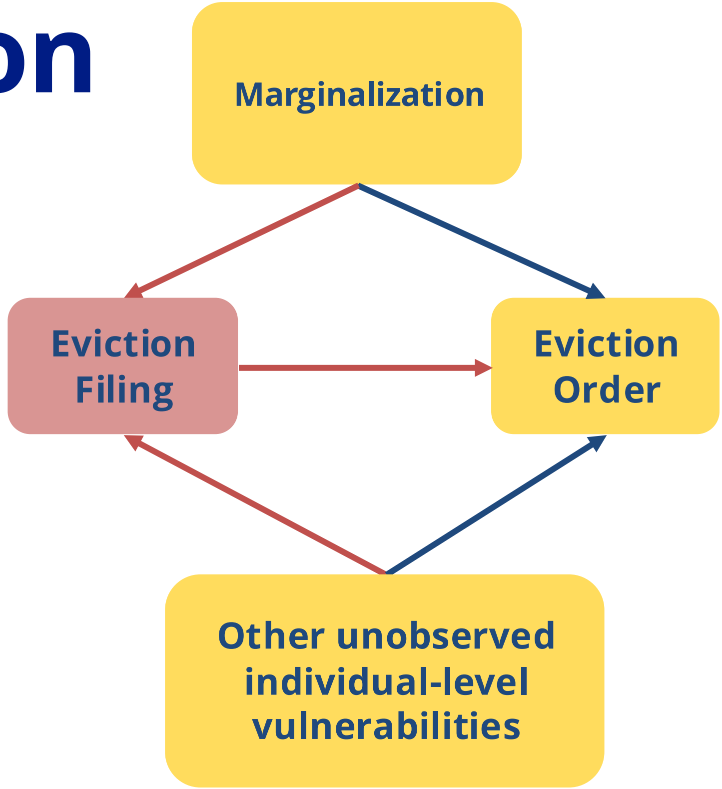
- … Black head of households are overrepresented in the population of people who receive an eviction filing

	All		Evicted		Not evicted	
Variables	*M* or %	*SD* or *N*	*M* or %	*SD* or *N*	*M* or %	*SD* or *N*
Total		19,748		8,404		11,344
Age	36.65	11.69	36.64	11.7	36.66	11.68
Prev. case in past year[a]	24.4%	3,438	25.8%	1,545	23.3%	1,888
Have children	59.5%	11,750	60.1%	5,051	59.1%	6,704
No. of children	2.147	1.27	2.17	1.30	2.13	1.26
Age under 25	14.9%	2,942	15.1%	1,269	14.8%	1,679
Age 25–39	48.9%	9,657	48.8%	4,101	49.0%	5,559
Age 40–59	32.2%	6,359	31.9%	2,681	32.4%	3,675
Age 60+	4.0%	786	4.2%	354	3.8%	432
Gender: female	78.4%	15,482	77.4%	6,505	79.2%	8,984
Gender: male	21.6%	4,266	22.6%	1,899	20.8%	2,360
Race: Black	76.6%	15,125	74.1%	6,227	78.4%	8,892
Race: Other, unknown	3.5%	691	3.4%	283	3.6%	409
Race: White	19.9%	3,929	22.5%	1,891	18.0%	2,042
Area deprivation index	88.59	12.85	88.55	12.6	88.61	13.02

Note. *SD* = standard deviation; *M* = mean; *N* = count. [a]Excludes 2013 filings to look back 1 year. Total eviction filings: 19,748.



Collider Bias in Eviction Research



- Without realizing that eviction filing is a **collider variable**, we may **inaccurately** interpret results as White individuals being more likely to receive an eviction order, thus concluding that **marginalization does not influence eviction outcomes**
- Using graphical models can help scrutinize our data and assumptions for colliders and **more accurately understand the data and the social system that underlies the data**

Note how the authors describe the study findings:

> Whereas Black or African American heads of household make up the majority of eviction filing cases, the likelihood of having an eviction move-out order is higher for White than for Black or African American heads of household. Similarly, women are much more likely to face an eviction filing, but among those with a filing, the risk of receiving an eviction order is higher for men than for women. The key to interpreting these apparently counterintuitive statistical patterns is to note that we are conditioning on individuals with an eviction filing, not the entire population. The relatively few low-income White males who receive an eviction filing may be individuals facing extreme struggles and are by no means a representative sample of the population of White males at large.

Richter, F. G. C., Coulton, C., Urban, A., & Steh, S. (2021). An Integrated Data System Lens Into Evictions and Their Effects. *Housing Policy Debate*, 31(3–5), 762–784. https://doi.org/10.1080/10511482.2021.1879201